Contents

1. Exploratory Data Analysis

This section presents an Exploratory Data Analysis (EDA) of the MVTec Anomaly Detection[1] dataset. MVTec AD is a benchmark dataset for unsupervised anomaly detection in industrial inspection and contains over 5,000 images across 15 object categories. Each category contains defect-free training images and a test set consisting of both normal and anomalous samples.

The purpose of this analysis is to:

1. Identify the object and texture categories present within the dataset.
2. Examine the distribution of normal and anomalous samples.
3. Investigate the defect types available within each category.
4. Quantify the spatial extent of anomalous regions using the provided segmentation masks
5. Analyse image characteristics such as resolution, brightness, sharpness, and contrast.
6. Identify potential challenges and limitations that may influence anomaly detection performance.
7. Inform preprocessing and model design decisions for the subsequent stages of the project.

1.1. Object and Texture Categories

All 15 available categories that are present within the MVTec dataset.

bottle	cable	capsule	carpet	grid
hazelnut	leather	metal_nut	pill	screw
tile	toothbrush	transistor	wood	zipper

1.2. Normal and Anomalous Sample Distribution

This section examines the overall composition of the dataset in terms of its train/test split and the proportion of normal and defective samples. These distributions provide context for how MVTec AD is structured for anomaly detection, with training data consisting of normal samples and anomalous examples appearing within the test set. Examining these proportions is important for interpreting later evaluation results, particularly because the available numbers of normal and defective test samples influence how category-level performance metrics are estimated.

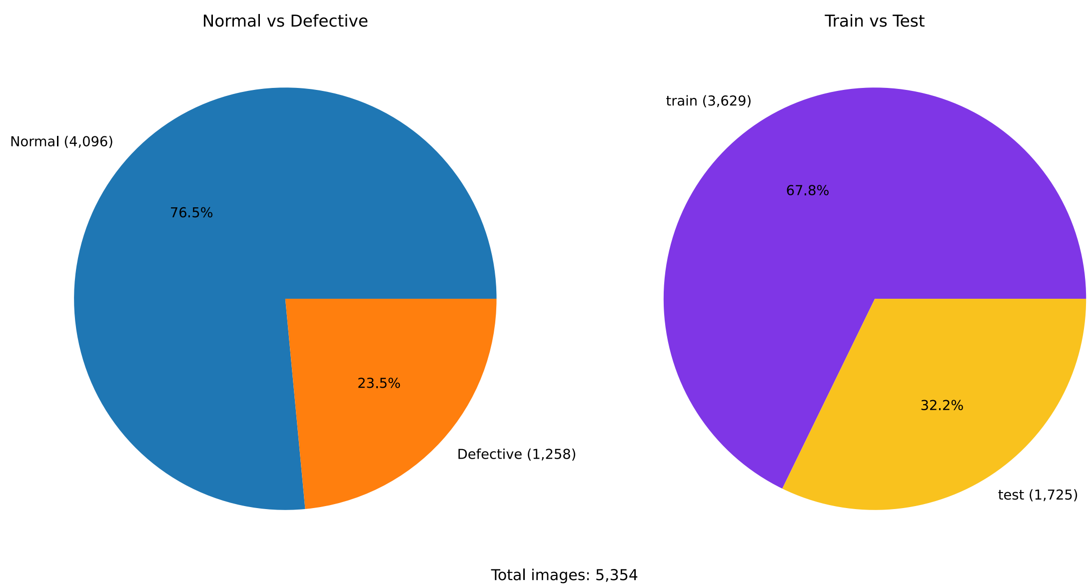


Figure 1: Overall normal/defective and train/test distribution of MVTec AD

1.2.1. Category Distribution

The distribution of images across the 15 MVTec AD categories is first examined to identify whether category-level imbalance could disproportionately influence the subsequent evaluation. As shown in Figure 2, the dataset is reasonably balanced across categories, although some variation in sample count is present. No individual category dominates the dataset sufficiently to suggest that the overall evaluation would be driven primarily by a single class.

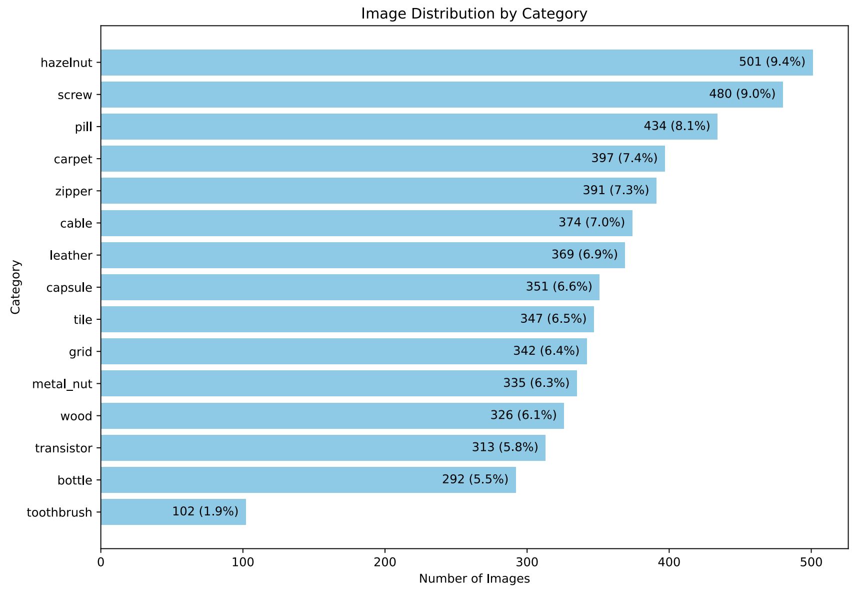


Figure 2: Distribution of images across MVTec AD categories

However, total category size does not describe how the available samples are distributed between normal and defective images. Figure 3 therefore separates the category distributions according to sample type. The proportion of defective samples varies between categories, meaning that the number of anomalous examples available for evaluation is not uniform across the dataset. This variation should be considered alongside model results, particularly where differences in cluster statistics or anomaly-detection performance are observed.

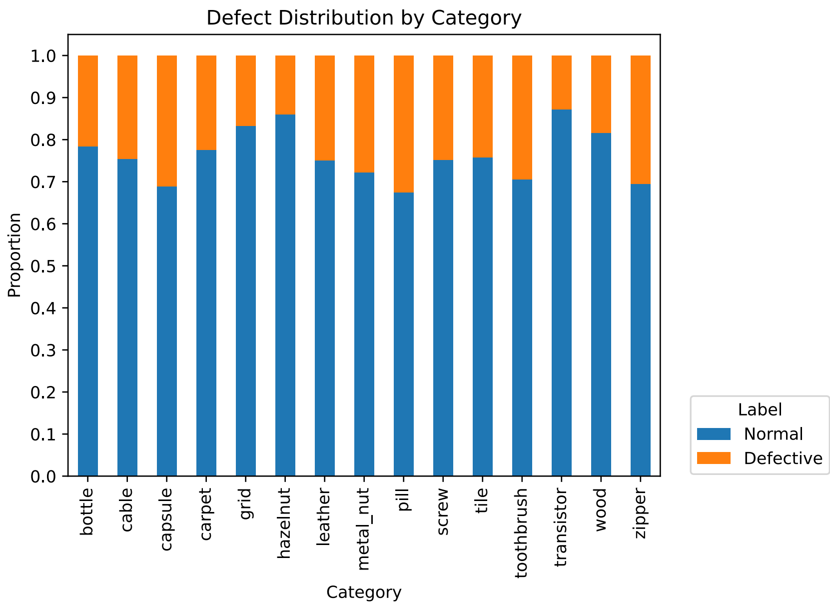


Figure 3: Proportion of normal and defective samples by category

Differences in the proportion of defective samples may also be related to the number and frequency of defect types present within each category. This is examined in the following section.

1.3. Defect Types by Category

The category-level distributions in Figure 3 show that the amount of defective data varies across MVTec AD. However, the number of defective samples alone does not describe the diversity of anomalies represented within each category. The individual defect types are therefore examined to determine how anomaly frequency and variation differ between categories.

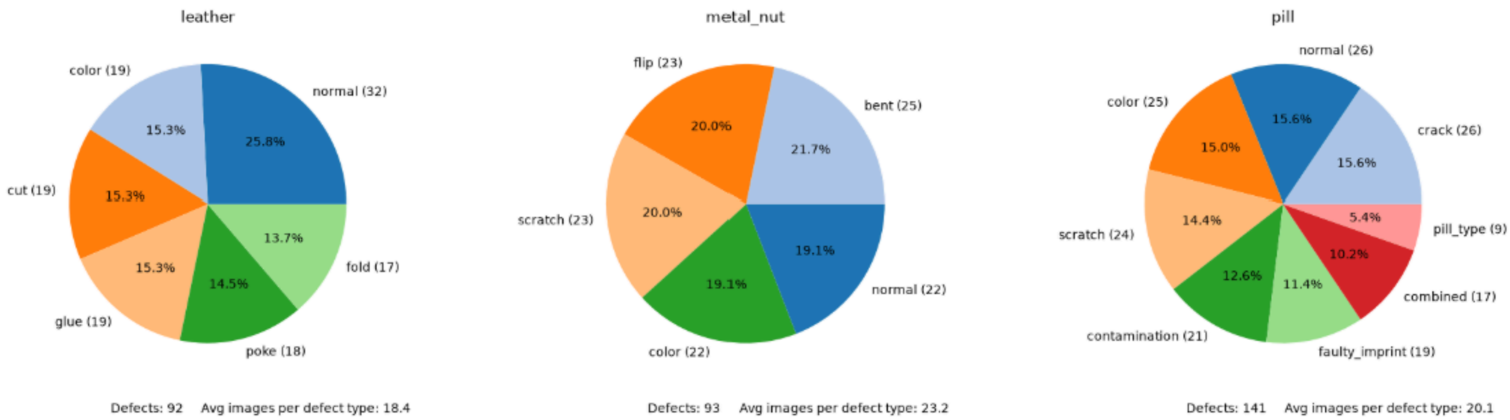


Figure 4: A sample of the distributions of defects in the data

Figure 4 shows examples of the distribution of defect types within the test sets. Considerable variation is present both in the number of defect types associated with each category and in their relative frequency. Some categories contain a small number of relatively common defect types, while others contain a wider range of anomalies with more uneven distributions. Consequently, two categories containing similar numbers of defective images may still represent substantially different anomaly-detection problems. For example, categories with a similar overall number of defective samples can still differ substantially in the composition of those defects. Leather contains five defect types, compared with four for metal nut, and the frequency of each defect type is not necessarily uniform. As a result, category-level statistics may be influenced more strongly by defect types that occur frequently, particularly if some anomalies are consistently easier or harder to distinguish than others. The number of normal test samples also varies between categories. While this does not inherently bias AUROC towards categories with more normal samples, smaller sample counts may produce less stable estimates of category-level performance. Consequently, both the diversity and distribution of defect types should be considered when interpreting differences between categories.

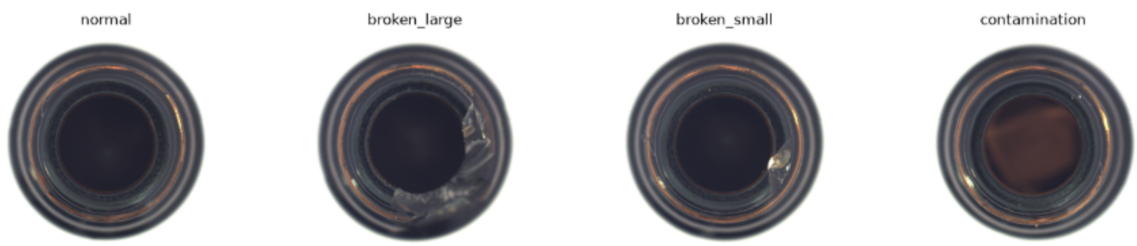
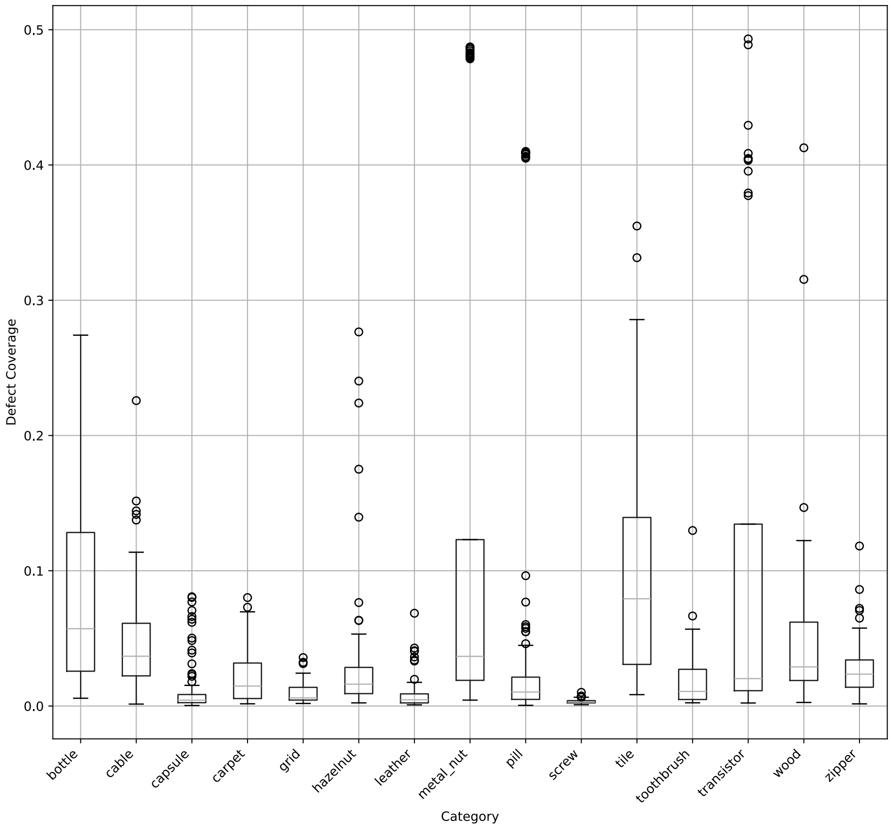


Figure 5: Example of defects in the bottle category

Example normal and defective images are also examined to provide visual context for these distributions. These examples illustrate that defect types differ not only in frequency but also in their appearance, spatial extent, and relationship to the underlying object or texture. This variation can be seen in Figure 5, where the defective samples do not all exhibit visually similar anomalies. Although some defects represent a similar underlying failure, such as broken_small and broken_large, they differ substantially in their spatial extent. Other defect types, such as contamination, can instead appear primarily as a local change in colour or surface appearance. Consequently, defect categories with similar sample counts may still contain anomalies that differ considerably in both scale and visual characteristics. These properties are examined further in the subsequent defect-coverage analysis and later embedding-space analysis, where their relationship with the learned representation is investigated.

1.3.1. Defect Size and Image Coverage

Defect severity is not determined solely by defect type, the spatial coverage of an anomaly may also influence detection difficulty. Using the provided segmentation masks, the proportion of each defective image occupied by anomalous regions was calculated to examine how defect size varies both within and between categories.



Category	Coverage
bottle	0.076
cable	0.047
capsule	0.011
carpet	0.021
grid	0.009
hazelnut	0.034
leather	0.009
metal_nut	0.145
pill	0.04
screw	0.003
tile	0.098
toothbrush	0.021
transistor	0.12
wood	0.051
zipper	0.026

Figure 6: Defect coverage by category accompanied by the average coverage by category.

As shown in Figure 6, defect coverage varies substantially both between and within categories. Metal nut is particularly notable: while many defective samples occupy approximately 10% of the image, several contain anomalous regions approaching 50%. As shown in Figure 4, one of the metal nut anomaly types consists of a flipped object, which may contribute to the substantially larger spatial coverage observed for some samples. Further inspection shows that these high-coverage samples correspond to the flip defect type, in which a substantial portion of the object is spatially altered. This explains the unusually large anomalous regions observed for the category and highlights how defect type can strongly influence spatial coverage. In contrast, screw contains particularly small anomalous regions, with defective samples covering only 0.3% of the image on average. Such limited spatial coverage may make this category more difficult to detect, particularly when using global image representations, where the anomalous region contributes only a small proportion of the overall representation.

1.4. Image Characteristics

This section investigates several image characteristics that may influence feature extraction and anomaly detection performance. Resolution, brightness, sharpness, and contrast are analysed to identify potential inconsistencies or preprocessing requirements.

Image-level statistics are computed for every image in the dataset. Brightness is measured as the mean pixel intensity, contrast as the standard deviation of pixel intensities, and sharpness using the variance of the Laplacian.

Category	**Brightness**		**Sharpness**		**Contrast**	
	mean	**std**	**mean**	**std**	**mean**	**std**
bottle	136.74	1.71	44.38	4.63	92.72	0.95
cable	102.06	4.71	46.01	10.6	50.77	2.81
capsule	172.24	4.32	17.25	3.37	63.0	1.68
carpet	91.06	2.48	300.72	56.27	35.39	1.54
grid	114.11	9.23	30.09	11.4	39.18	7.52
hazelnut	51.72	4.13	43.01	3.84	26.49	5.13
leather	80.39	6.91	46.96	18.76	11.75	2.09
metal_nut	58.49	4.39	62.75	14.18	43.25	4.58
pill	76.33	4.96	142.1	96.86	78.45	4.05
screw	183.61	6.35	23.41	2.13	33.42	1.98
tile	116.35	8.45	46.61	14.97	31.5	2.23
toothbrush	47.02	8.02	167.17	137.77	55.81	1.61
transistor	79.52	3.44	63.58	8.43	42.07	1.56
wood	135.9	7.2	89.45	24.39	16.09	3.04
zipper	101.97	2.02	72.46	9.66	82.33	1.23

Table 1: Image-level statistics by MVTec AD category.

1.4.1. Observations

Several notable observations can be made from the image characteristic analysis:

- Image resolution is consistent within each category but varies between categories, ranging from 700x700 to 1024x1024 pixels.
- Brightness varies significantly between categories due to differences in object appearance and material properties, but remains highly consistent within individual categories.
- Sharpness and contrast exhibit greater variation between categories, reflecting differences in texture complexity and surface structure.
- Categories such as carpet, wood, and toothbrush contain substantially more texture detail than smoother categories such as capsule and screw.

These findings suggest that resizing and normalization will be necessary during preprocessing, while category-specific texture complexity may influence anomaly detection performance.

2. Baseline Model

This section explores the pretrained embeddings produced by the DINOv2 ViT-S/14 model. DINOv2 was selected because its self-supervised training produces general-purpose visual representations without requiring task-specific labels, making it well suited to the unsupervised setting considered in this work. The ViT-S/14 (approximately 21 million parameters) variant was chosen as a relatively compact model that provides both global CLS and local patch representations, allowing the same backbone to support the image-level experiments in this study and subsequent investigation of local anomaly representations.

The purpose of this section is to:

1. Evaluate several anomaly scoring methods using the extracted CLS embeddings:
   - Centroid Distance
   - K-Nearest Neighbours (KNN)
   - Mahalanobis Distance
2. Analyse the distribution of anomaly scores for normal and defective samples to determine how well each method separates the two classes.
3. Compare methods using AUROC to identify the most effective baseline anomaly scoring approach. AUROC is used because it evaluates ranking performance across all possible decision thresholds, avoiding the need to select a fixed anomaly threshold for baseline comparison.
4. Identify categories that are already well separated within the embedding space, as well as categories that remain challenging and may benefit from additional experimentation.

The results obtained in this section serve as a baseline against which subsequent approaches, including fine-tuned and semi-supervised representations, are compared.

2.1. Evaluate several anomaly scoring methods using the extracted CLS embeddings

Metrics chosen include:

- **Centroid Distance:** Computes the Euclidean distance between each test embedding and the centroid (mean embedding) of the corresponding training category.
- **KNN:** The training embeddings are used as the reference set for KNN. Each test embedding is then compared against them to find its K nearest training embeddings and their distances. The anomaly score is taken as the distance to the closest neighbour.
- **Average KNN:** Computes the average distance between each test embedding and its K (K=5) nearest training embeddings.
- **Mahalanobis Distance:** Computes the distance between each test embedding and the training distribution while accounting for the covariance structure of the embedding space.

2.2. Analyse the distribution of Anomaly Scores

The anomaly score distributions of normal and defective test samples are compared separately for each category and scoring method to examine how effectively the pretrained CLS embeddings distinguish anomalous samples from the learned normal representation. AUROC is computed for each category to quantify how consistently defective samples receive higher anomaly scores than normal samples. The methods are first compared using their aggregate performance across all categories, after which the category-level results of the selected Mahalanobis baseline are examined in greater detail. These results provide a reference for the subsequent embedding-space analysis, where differences in representation structure are considered in relation to anomaly-detection performance.

Method	Mean	Median	Std	Min Category	Value	Max Category	Value
Centroid	0.845	0.849	0.135	screw	0.506	leather	0.997
KNN	0.923	0.933	0.06	screw	0.804	leather	1.0
Avg-KNN	0.92	0.926	0.062	screw	0.786	leather	1.0
Mahalanobis	0.958	0.972	0.049	screw	0.802	tile	1.0

Table 2: AUROCs for each scoring method

The centroid-based method has reasonable performance across several categories; however, there is a noticeable overlap between the score distributions of normal and defective samples. This indicates that the distance from the centroid alone is not enough to fully describe the embedding space. While the centroid gets the average location of normal samples, it does not take into account the spread or covariance of the distribution, meaning some anomalous samples receive scores similar to normal samples.

The K-Nearest Neighbour approach improves upon the centroid-based method by producing clearer separation between the score distributions of normal and defective samples. The leather category has perfect separation, achieving an AUROC of 1.0. This suggests that anomalous samples are characterised more by their distance to nearby normal samples than by their distance to a single global centroid. The results indicate that normal and defective samples occupy distinct regions of the embedding space, allowing KNN to identify anomalies through local neighbourhood structure rather than relying solely on the overall centre of the distribution.

Average KNN does not improve upon the performance of standard KNN and achieves a lower average AUROC across the dataset. This suggests that the nearest normal neighbour contains the most informative signal for anomaly detection. By averaging across multiple neighbours, the anomaly score becomes influenced by additional normal samples that may lie further away in the embedding space, reducing the contrast between normal and defective samples. As a result, the score distributions exhibit greater overlap, leading to weaker separation and reduced anomaly detection performance.

The Mahalanobis score performed best among all evaluated methods, achieving perfect separation between normal and defective samples in several categories, including bottle, leather, and tile. Mahalanobis achieved the highest mean AUROC overall. KNN performed marginally better on screw, achieving 0.804 compared with 0.802 for Mahalanobis. Overall, Mahalanobis was therefore selected as the preferred anomaly scoring method for this study. Furthermore, the categories exhibiting perfect separation could be used as control groups in future experiments to evaluate whether proposed modifications improve anomaly detection performance or instead degrade the structure and separability of the embedding space.

Despite strong overall performance, categories such as screw achieved lower AUROC values across multiple scoring methods. This suggests that normal and defective samples occupy more similar regions of the embedding space, making anomaly detection more challenging. These categories provide useful benchmarks for evaluating future improvements, as successful methods should increase separability in these difficult cases.

2.2.1. Anomaly scores across categories

The category-level Mahalanobis results provide additional context to the aggregate results, where tile was identified as the best-performing category with an AUROC of 1.0. The full category breakdown shows that this performance is not unique to tile, with bottle and leather also achieving complete normal-defective separation. These three categories therefore provide useful reference cases for subsequent experimentation, as their embedding characteristics can be compared with more challenging categories to investigate which properties are associated with strong anomaly separation.

Mahalanobis Across Categories

Category	Mahalanobis
tile	1.0
zipper	0.987
toothbrush	0.972
hazelnut	0.949
wood	0.944
grid	0.983
cable	0.945
leather	1.0
metal_nut	0.983
screw	0.802
bottle	1.0
pill	0.95
capsule	0.941
carpet	0.98
transistor	0.935

Table 3

Another potentially interesting comparison can be made between categories with relatively uniform, full-frame appearances, such as tile, wood, leather, and carpet. Although tile and leather achieve perfect Mahalanobis separation, wood and carpet do not. This raises the question of which characteristics of visually uniform categories contribute to stronger anomaly separation. Defect coverage is one possible factor; however, the values in Figure 6 do not suggest a simple relationship. The perfectly separated categories, leather and tile, have average defect coverages of approximately 0.9% and 9.8%, respectively, while carpet and wood have intermediate coverages of approximately 2.1% and 5.1%. This suggests that defect extent alone is unlikely to explain the difference in AUROC. Other factors may include how visually distinctive a defect is from the surrounding texture, whether the anomaly produces a strong local contrast, or how the pretrained representation responds to different spatial regions of the image. Differences in the visual characteristics represented during pretraining could also contribute, although this cannot be directly verified from the current analysis.

3. Embedding Space Visualisation

This section investigates the structure of the pretrained embedding space to determine what information is represented before task-specific training and how this structure relates to anomaly detection performance. Analysis is performed at both the global image level, using CLS embeddings, and the local level, using patch embeddings. This distinction allows the representation of complete images to be compared with the local structure associated with individual anomalous regions, providing insight into whether anomaly information that is distinct at the patch level is retained within the final global representation.

Embedding structure is examined qualitatively using PCA, t-SNE, and UMAP, and quantitatively using measures of cluster separation, compactness, and distance from the learned normal representation. These analyses are used to investigate differences between normal and defective samples, variation between dataset categories, and whether characteristics identified during the earlier EDA, such as defect coverage and visual structure, help explain the observed behaviour. The resulting embedding characteristics are also considered in relation to the baseline AUROC results to identify properties that may contribute to strong or weak anomaly separation.

Scoring metrics will be used to analyse cluster quality and separation, including:

- **Silhouette Score (Sil.)**
  Measures how similar samples are to their own cluster relative to the opposing cluster. Higher values indicate stronger separation, with a maximum value of 1. This metric is used to assess how distinctly the pretrained embedding space separates normal and defective representations.
- **Davies-Bouldin Index (DB)**
  Measures within-cluster dispersion relative to the separation between clusters. Lower values indicate more compact and well-separated clusters. This provides a complementary measure to the Silhouette Score by considering how tightly normal and defective embeddings are grouped relative to the distance between them.
- **Calinski-Harabasz Index (CH)**
  Measures between-cluster separation relative to within-cluster variation, with higher values indicating stronger cluster structure. This is used to determine whether differences between normal and defective representations are large relative to the variability present within each group.
- **Intra-Cluster Distance**
  Measures how far embeddings within a group lie from their own centroid, where a value of 0 would indicate that all embeddings occupy the same point. This is used to examine the compactness and variability of normal and defective representations, and at the patch level also allows the stability of normal representations between the training and test distributions to be compared.
- **Inter-Cluster Distance**
  Measures the distance of embeddings from a shared reference centroid. In this analysis, the normal training representation is generally used as the reference, allowing the distances of normal and defective test embeddings to be compared. This is used to determine whether defective representations move farther from the learned normal distribution and therefore contain a useful distance-based anomaly signal.

Together, these metrics provide complementary views of the embedding space: Silhouette, DB, and CH evaluate normal-defective cluster structure, while the intra- and inter-cluster distances examine the compactness of individual representations and their displacement from the learned normal reference.

3.1. CLS Embeddings

The CLS embedding provides a single global representation of an input image. In the Vision Transformer, information from the image patches is aggregated into the CLS token, producing one embedding that represents the image as a whole. In this work, the CLS representation is used for image-level anomaly detection, where normal and defective samples are compared within the pretrained embedding space. The following subsections first examine the organisation of these embeddings visually and then quantify their normal-defective structure using cluster and distance-based statistics.

3.1.1. Embedding Projections

First establish how DINOv2 organises the 15 visual categories globally. Then examine whether normal and defective samples separate within those category structures.

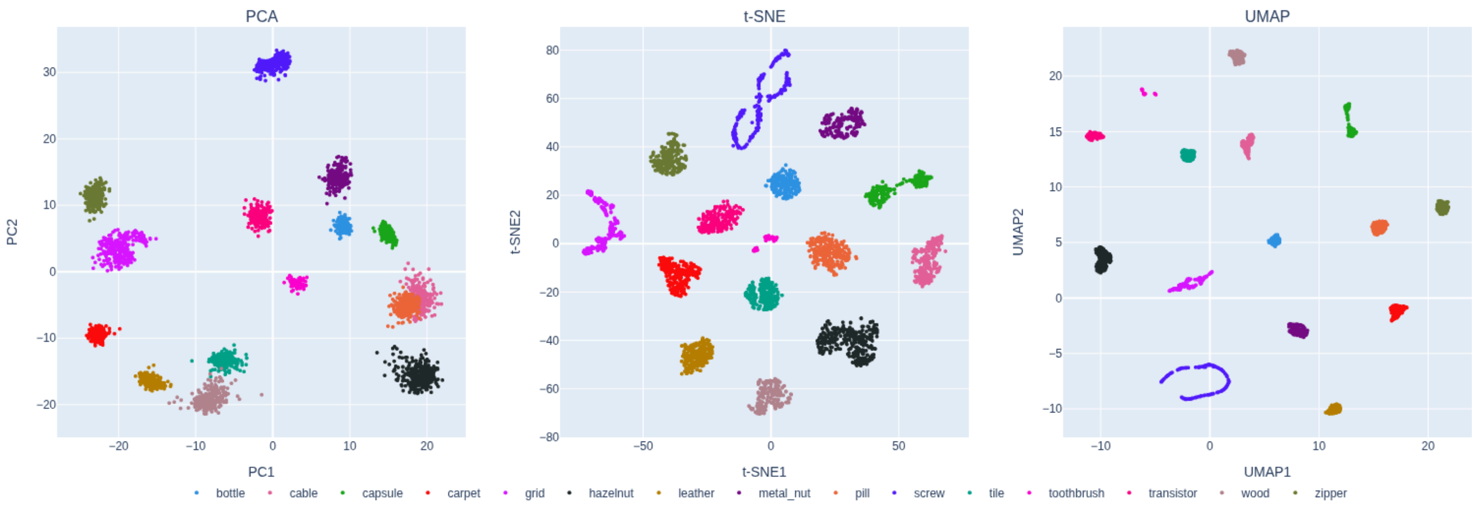


Figure 7: Embedding Projections of CLS tokens

Within the PCA projection, we can observe how the embedding space is organised along the two principal components that capture the greatest variance in the data. Several categories form distinct clusters that reflect similarities in their visual appearance. For example, relatively flat and homogeneous objects such as tile, leather, and wood are positioned closer together, while categories such as screw occupy a more distant region of the embedding space due to their more complex geometric structure and visual texture.

In the t-SNE projection, we can observe how compact or diffuse each category is within the embedding space. Most categories form relatively tight clusters, indicating that samples from the same category are represented consistently by the model. Notable exceptions include grid and screw, which exhibit more diffuse and fragmented structures. The screw category is particularly interesting, as it also achieved some of the lowest anomaly detection performance in the previous analysis. Rather than forming a single compact cluster, the embeddings are distributed in a figure-eight like structure, suggesting that normal and defective samples may occupy multiple overlapping regions of the embedding space. This reduced compactness may contribute to the lower AUROC scores observed for this category, as anomalous samples become more difficult to distinguish from normal samples using anomaly scoring methods.

Finally, the UMAP projection highlights how distinct the local neighbourhoods are within the embedding space and how compact these neighbourhoods remain. The observations regarding cluster compactness are similar to those seen in the t-SNE projection, with grid and screw again emerging as the most diffuse categories. Rather than forming a single coherent cluster, samples from these categories occupy multiple regions of the embedding space, with the screw category appearing to split into three distinct subregions. This provides further insight into the lower anomaly detection performance observed for screw in the previous analysis. Since normal samples are distributed across several disconnected regions rather than a single compact cluster, anomaly scoring methods may struggle to distinguish normal and defective samples consistently, resulting in reduced AUROC performance.

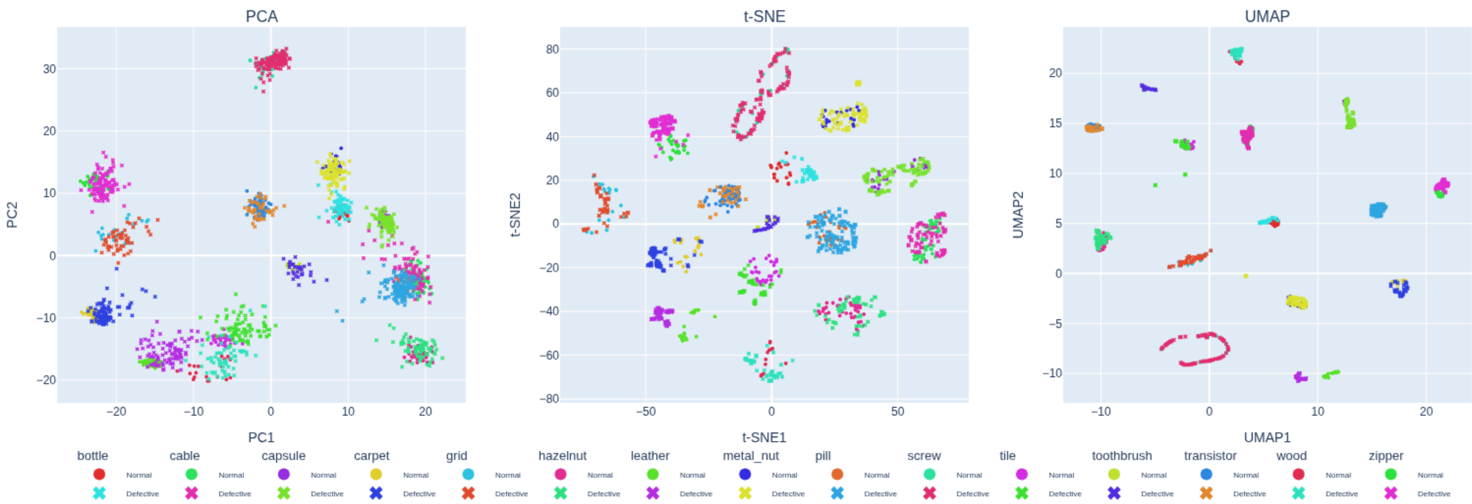


Figure 8: Embedding Projections of Normal vs Defective CLS Tokens

For the PCA projection, there is unsurprisingly little separation between normal and defective samples. Since PCA preserves the directions of greatest global variance, the principal components are dominated by category-level differences rather than the subtle differences introduced by anomalies. As a result, normal and defective samples largely occupy the same regions of the embedding space.

Within the t-SNE projection, we can observe how defective samples are positioned relative to local neighbourhoods of normal samples. For many categories, particularly those containing flat and visually consistent objects, normal and defective samples form relatively distinct regions. However, the screw category remains an exception. Here, normal and defective samples occupy almost identical regions of the embedding space, with little evidence of a distinct anomaly cluster. This provides a possible explanation for the lower anomaly detection performance observed previously, as the embedding representation does not naturally separate normal and defective screw samples.

The UMAP projection exhibits similar behaviour to t-SNE. For many categories, defective samples tend to occupy regions adjacent to, but partially separated from, normal samples. However, in more challenging categories such as screw, normal and defective samples continue to share the same embedding regions. This suggests that the model struggles to learn discriminative representations for these anomalies, making them difficult to separate using distance-based anomaly scoring methods.

3.1.2. Embedding Statistics

This section quantitatively evaluates the structure of the CLS embedding space identified in the preceding projections. The analysis examines normal-defective cluster quality, the position of test embeddings relative to the learned normal training representation, and the compactness of normal and defective groups. These statistics are used to determine whether the visual structures observed in the projections are also present in the original embedding space and to investigate how differences in CLS geometry relate to dataset characteristics and the baseline anomaly-detection results.

Metric	Mean	Median	Std	Min	Max	Worst	Best
Cluster Quality							
Sil.	0.0401	0.0425	0.0588	-0.0722	0.1615	capsule	leather
DB	3.2804	2.8931	1.4754	1.5058	6.6545	screw	leather
CH	8.8111	6.6562	6.7326	2.6555	27.5506	screw	leather
Inter-Cluster Normal-Reference Distances							
Normal	13.5318	12.1777	4.0829	7.3152	21.3277	wood	bottle
Defect	20.5365	19.4394	5.0035	13.0549	30.4626	capsule	tile
Sep. Ratio	1.5956	1.3713	0.45	0.9871	2.4991	screw	bottle
Intra-Cluster Distance							
Normal	12.6352	11.4615	3.9479	6.9367	18.9459	wood	bottle
Defect	17.7923	16.6142	3.7904	12.5704	25.5036	capsule	tile

Table 4: CLS Embedding Statistics

The mean Silhouette Score across categories is 0.04, indicating weak separation between normal and defective CLS embeddings. Since a score close to zero indicates substantial overlap between clusters, the result suggests that normal and defective samples often occupy similar regions of the CLS embedding space. This can be seen for the best-performing (leather: 0.1615) and worst-performing (capsule: −0.0722) categories in Figure 8. For leather, the normal and defective embeddings are comparatively well separated in the projection, forming two more distinct regions. In contrast, capsule shows substantial overlap between normal and defective samples, with both occupying similar regions of the projected embedding space. However, this difference is not explained simply by the spatial extent of the anomalies. As shown in Figure 6, defects occupy approximately 0.9% of the image area on average for leather and 1.1% for capsule, despite their substantially different Silhouette Scores. The defect categories shown in Figure 4 indicate that the leather class contains several distinct anomaly types, suggesting that characteristics other than spatial extent may influence how defects are represented. Further analysis of the individual defect types would be required to determine whether these differences contribute to the relatively strong CLS separability observed for leather.

The mean Davies-Bouldin Index across categories is approximately 3.28, indicating that the normal and defective CLS embeddings generally exhibit substantial within-cluster dispersion relative to the distance separating their clusters. This is most apparent for the worst-performing category, screw (6.6545), where Figure 8 shows that neither the normal nor defective samples form clearly distinct regions, with both distributed broadly across the projected embedding space. In contrast, the best-performing category, leather (1.5058), forms more clearly separated normal and defective regions in Figure 8, where this strong CLS structure is also reflected in Table 3 results, where leather achieves an AUROC of 1.0. However, both regions remain relatively dispersed rather than forming highly compact clusters, which is consistent with a DB value that, although substantially lower than screw and the overall mean, is still above zero. As these observations are based on a low-dimensional projection of the original embedding space, the projection alone cannot fully explain the DB values and should be treated as supporting visual evidence rather than a direct representation of the full embedding geometry.

The mean Calinski-Harabasz Index across categories is 8.8111, substantially lower than the best-performing category, leather (27.5506). The comparatively high CH value for leather indicates that the separation between its normal and defective CLS embeddings is large relative to the variation within those groups. This is consistent with the preceding Silhouette analysis, where leather also exhibited the strongest normal-defective separation and where its relatively strong separability was discussed in relation to the defect characteristics shown in Figure 4 and Figure 6. In contrast, screw achieves the lowest CH value (2.6555), indicating that the separation between its normal and defective embeddings is small relative to their within-group variation. This may relate to the defect coverage reported in Figure 6, where anomalous regions occupy only approximately 0.3% of the image on average, potentially limiting their influence on the global CLS representation.

Both the Normal and Defect Inter-Cluster Distances are measured relative to the centroid of the normal training embeddings for each category, rather than relative to one another. Across categories, normal samples have a mean distance of 13.53 from this reference, compared with 20.54 for defective samples, producing a mean Separation Ratio of 1.60. This indicates that defective CLS embeddings are generally positioned farther from the learned normal representation. However, the strength of this distinction varies considerably between categories. Bottle achieves the highest Separation Ratio of 2.4991, indicating that its defective embeddings lie approximately 2.5 times farther from the normal training centroid than its normal test embeddings. This is consistent with its perfect AUROC of 1.0 from Table 2, suggesting that its strong displacement from the normal reference is reflected in the resulting anomaly score. In contrast, screw achieves the lowest ratio of 0.9871, meaning that its defective samples are, on average, marginally closer to the normal training centroid than its normal test samples. This is particularly problematic for distance-based anomaly scoring, since distance from the learned normal representation provides little discriminatory signal for this category, which this behaviour can be seen back to screw consistently being the lowest category in Table 3. This is also consistent with the high normal-defective overlap previously observed for screw in Figure 8. The variation also motivates investigating whether category-level Separation Ratio is associated with anomaly-detection performance by comparing it with AUROC.

Normal Intra-Cluster Distance is generally lower than Defect Intra-Cluster Distance, indicating that normal samples form more compact clusters while defective samples exhibit greater variability. This may partly reflect the diversity of defect types within each category, since different anomalies can alter the visual representation in different ways. Consequently, defective CLS embeddings may occupy multiple distinct regions or neighbourhoods rather than forming a single coherent cluster, increasing their average distance from the defective centroid. The earlier Figure 6 also showed substantial variation in defect coverage within some categories, suggesting that defective samples can differ considerably in both appearance and spatial extent. This additional variation may further contribute to the greater spread of defective embeddings around their centroid.

3.2. Patch Embeddings

Patch embeddings retain local representations for individual image regions rather than collapsing the image into a single global embedding. This allows anomalous regions to be examined independently of the surrounding normal content and provides a way to investigate whether local anomaly information is more distinct than it appears in the CLS representation. The following subsections quantify patch-level structure and then examine selected defect types visually.

3.2.1. Embedding Statistics

The patch-level Silhouette Scores are consistently positive and generally higher than the mean CLS score of approximately 0.04 reported in Table 4, suggesting that local patch representations contain clearer normal-defective structure than the global image representation. However, the two values should not be interpreted as directly equivalent. Whereas the CLS score evaluates normal and defective images collectively within each category, the patch-level score is calculated separately within each defective image by comparing its normal and anomalous patches, before being averaged across the category. This means that local anomalous regions are evaluated within the context of their own image, rather than requiring visually different defect types to occupy a common region of the category-level embedding space. The stronger patch-level separation therefore provides additional support for the earlier CLS observation that anomaly characteristics, rather than spatial extent alone, may influence how clearly defects are represented.

Cluster Quality Metrics

Category	Sil.	DB	CH
bottle	0.1032	2.0161	21.0375
cable	0.1234	2.0764	17.4583
capsule	0.1103	1.4781	8.0617
carpet	0.2666	1.1308	25.0733
grid	0.2727	1.1495	22.83
hazelnut	0.1479	1.4992	19.9355
leather	0.2748	0.946	14.5154
metal_nut	0.0924	2.0189	31.2216
pill	0.0977	1.7748	14.5052
screw	0.1252	1.2913	5.5748
tile	0.2694	1.4006	56.9894
toothbrush	0.106	1.7726	13.9147
transistor	0.086	2.7955	13.6462
wood	0.2768	1.4002	40.1426
zipper	0.083	1.7521	12.1111

Table 5a

This interpretation is also supported by the EDA. As shown in Figure 4 and Figure 6, categories such as leather and grid achieve some of the highest patch-level Silhouette Scores despite containing relatively small anomalous regions, while metal nut exhibits substantially greater average defect coverage but a comparatively low Silhouette Score. This suggests that patch-level separability depends more strongly on the visual characteristics and consistency of the anomaly than on its spatial extent alone. Because anomalous regions usually occupy only a small part of each image, the patch-level Silhouette Score may be influenced by the much larger number of normal patches. However, differences in Silhouette Score do not appear to follow category sample count directly, as shown in Figure 6. This suggests that the variation between categories cannot be explained simply by differences in the number of available samples.

The mean patch-level Davies-Bouldin Index across categories is approximately 1.63, considerably lower than the mean CLS value of 3.28 reported in Table 4. This suggests that, overall, the normal and anomalous patch embeddings exhibit lower within-cluster dispersion relative to the separation between them than the corresponding CLS representations. The relatively low DB values observed for categories such as leather, carpet, and grid support the stronger Silhouette Scores, indicating that their normal and defective patch embeddings form comparatively compact and separated local structures. This is particularly notable for grid, which appeared diffuse in the earlier CLS projections Figure 7, suggesting that local patch representations can retain coherent anomaly-related structure even where the global image representation is more fragmented. Compared with Table 4, leather remains the best-performing category, with its DB value decreasing from 1.5058 at the CLS level to 0.946 at the patch level. However, the behaviour of screw changes substantially: despite having the worst CLS DB value of 6.6545, its patch-level DB falls to 1.2913, below the patch-level mean. This suggests that the patch representation captures more coherent local structure for screw than is apparent in its global CLS representation. Whether these changes in embedding structure are associated with anomaly-detection performance is can be for future experimentation.

The mean patch-level Calinski-Harabasz Index is approximately 21.13, compared with 8.81 for the CLS embeddings reported in Table 4, indicating stronger between-group separation relative to within-group variation at the patch level. However, this structure varies considerably between categories. Screw remains the worst-performing category, with a patch-level CH value of 5.5748 compared with 2.6555 at the CLS level. As shown in Figure 6, screw defects occupy only approximately 0.3% of the image on average, meaning that relatively few patches contain anomalous information. This suggests that defect coverage may influence patch-level cluster structure by limiting the amount of anomalous information available for separation. In contrast, tile, with an average defect coverage of approximately 9.8%, achieves the highest patch-level CH value of 56.9894. However, defect coverage does not fully explain the variation between categories, indicating that characteristics of the anomalies themselves also influence the resulting patch embedding structure.

Inter-Cluster Distances

Category	Normal	Defect	Sep. Ratio
bottle	35.567	38.7165	1.0888
cable	36.312	42.2481	1.1644
capsule	37.6489	45.1315	1.1989
carpet	28.0213	44.2434	1.5799
grid	29.8463	44.9653	1.5073
hazelnut	39.1841	47.6354	1.2157
leather	30.8956	49.6409	1.6115
metal_nut	37.5273	37.4852	0.9997
pill	37.9695	42.7971	1.1274
screw	37.5862	45.9468	1.2225
tile	32.41	45.3849	1.3964
toothbrush	37.6068	43.5274	1.1573
transistor	32.5668	37.1644	1.1422
wood	30.474	42.6916	1.4054
zipper	33.2571	37.3918	1.1236

Table 5b

For each category, a normal reference centroid is constructed from the patch embeddings of all normal training images. Both Normal and Defect Inter-Cluster Distances are measured relative to this shared training centroid for each test image, before being averaged across the category. Across categories, the mean Normal Inter-Cluster Distance is approximately 34.46. Several categories exhibit comparatively low normal-reference distances, including carpet (28.0213), grid (29.8463), wood (30.4740), and leather (30.8956), indicating that their normal patches remain relatively close to the learned normal reference. For categories such as carpet, leather, and wood, this may reflect comparatively consistent local appearance. Grid is particularly interesting because it also achieves one of the strongest patch-level Silhouette Scores. One possible explanation is its spatial structure,much of each image consists of background between relatively thin grid elements, meaning that many patches may contain highly similar normal content, while patches intersecting a defect can contain more distinctive local changes. This could contribute both to the relatively small normal-reference distance and the stronger separation observed for anomalous patches. However, further patch-level analysis would be required to determine whether this spatial structure is responsible for the observed behaviour.

Defect Inter-Cluster Distance averages approximately 43.00 across categories. A notable contrast with the CLS results in Table 4 is observed for capsule. At the CLS level, capsule has the lowest distance (13.0549), whereas its patch-level distance of 45.1315 is above the patch-level mean. As shown in "eda_table", capsule defects occupy only approximately 1.1% of the image on average. This suggests that anomalous regions can be locally distinct within the patch embedding space while contributing relatively little to the global CLS representation because they occupy only a small proportion of the image. This provides further evidence that local anomaly information may be weakened when aggregated into a global image representation.

Most categories achieve Separation Ratios above 1, with a mean of approximately 1.263, indicating that defective patches generally lie farther from the normal reference than normal patches. Leather achieves the strongest separation (1.6115), while metal nut has a ratio of approximately 1.00 (0.9997), indicating almost no difference between the average normal and defective patch distances. This result is particularly interesting in relation to Figure 6, where metal nut has the largest average defect coverage, and the earlier EDA identified the flip defect type as contributing to this unusually large anomalous area. A possible explanation is that a spatially large defect does not necessarily produce locally distinctive patch content. In the case of a structural anomaly such as a flipped object, many patches may still contain visual features similar to those observed in normal samples despite their altered global arrangement. This could reduce the distinction between normal and defective patch distances and provides further evidence that defect coverage alone does not determine the quality of the learned anomaly representation.

Another notable contrast is bottle, which achieves a relatively low patch-level Separation Ratio of 1.0888 despite having the strongest CLS-level Separation Ratio of 2.4991 in Table 4. Conversely, grid exhibits an above-average Defect Inter-Cluster Distance and one of the highest patch-level Separation Ratios (1.5073), despite showing substantially weaker structure at the CLS level. These contrasting behaviours indicate that strong local anomaly separation does not necessarily translate directly into strong global separation, and vice versa. This suggests that the relationship between the number and spatial distribution of anomalous patches, their local distinctiveness, and the extent to which this information contributes to the global CLS representation warrants further investigation.

Similarly to the inter-cluster analysis, the normal patch intra-cluster distances show substantial variation between categories, with a mean of approximately 31.55. Some of the most compact normal patch distributions are observed for carpet (24.8757), grid (24.6649), wood (23.5885), and leather (26.4941). This may partly reflect the consistent local appearance of these categories, as discussed in the preceding inter-cluster analysis. For grid, the large regions of visually similar background between the thin grid structures may also contribute to the comparatively compact normal patch representation. Normal test-patch compactness is additionally compared with that of the normal training patches to determine whether the spread of normal representations remains consistent when they occur within defective test images. On average, the normal test patches are slightly more compact than the training patches, with mean intra-cluster distances of approximately 31.55 and 32.85, respectively. In defective images, some categories contain fewer normal patches because a larger proportion of the image is occupied by anomalous regions. This can make the estimated test intra-cluster distance less stable for those images, although it does not inherently cause the distance itself to be lower.

Intra-Cluster Distances

Category	Normal	Train	Ratio	Defect
bottle	33.8505	34.6712	0.9763	24.6126
cable	34.2263	35.7712	0.9568	27.7112
capsule	37.0514	37.4239	0.99	20.1856
carpet	24.8757	25.285	0.9838	15.8434
grid	24.6649	27.4793	0.8976	16.2601
hazelnut	37.6479	38.9374	0.9669	23.3492
leather	26.4941	24.4616	1.0831	12.5874
metal_nut	34.3597	36.5529	0.94	25.8242
pill	36.554	37.7873	0.9674	23.9242
screw	35.9715	37.1229	0.969	17.2762
tile	25.9535	26.6325	0.9745	22.2262
toothbrush	36.3193	37.0921	0.9792	25.1748
transistor	30.3868	32.6993	0.9293	27.0359
wood	23.5885	28.6674	0.8228	20.3072
zipper	31.261	32.1882	0.9712	19.5223

Table 5c

Most categories have a Normal Test/Train ratio close to 1, with a mean of approximately 0.961, suggesting that the compactness of normal patches is generally preserved between the training and test distributions. However, some categories deviate more substantially. Wood achieves the lowest ratio of 0.8228, indicating the largest reduction in normal patch dispersion relative to its training distribution. This behaviour cannot be explained straightforwardly by defect coverage alone. As shown in Figure 6, wood has an average defect coverage of approximately 5.1%, while bottle has a larger average coverage of 7.6% but retains a ratio much closer to 1 (0.9763). Interestingly, wood also has the highest Normal Intra-Cluster Distance at the CLS level (18.9459) in Table 4, indicating comparatively high dispersion in its global normal representation. The combination of a large change in patch-level compactness and high CLS-level dispersion suggests that the relationship between local patch structure and the resulting global representation warrants further investigation. The following section examines these embedding characteristics further, while future work could investigate how local patch information is aggregated into the CLS token.

Defect Intra-Cluster Distance is particularly interesting, as defective patch embeddings are substantially more compact than either the normal test or normal training patch distributions, with a mean distance of approximately 21.46. This suggests that anomalous patches within individual defective images often form relatively coherent local structures in the embedding space. One possible explanation is that patches containing the same anomaly share strongly related visual information, causing their representations to occupy similar neighbourhoods. However, the current distance-based analysis cannot determine how this structure arises within the transformer. Future investigation could examine the attention between anomalous patch tokens, their interaction with surrounding normal patches, and the extent to which this local information is subsequently propagated into the CLS representation. This could help determine whether the compactness observed at the patch level is related to how anomalous information is aggregated by the model.

3.2.2. Embedding Projections

The preceding patch statistics describe category-level behaviour by aggregating measurements across defective images. However, individual defect types can differ substantially in their spatial extent and visual characteristics. This section therefore examines selected patch projections at the defect-type level to provide qualitative context for the statistical results. Screw, metal nut, and wood are considered as case studies because they exhibit different behaviours in the preceding analysis, allowing local anomaly structure, defect coverage, and variation in normal patch representations to be examined more directly.

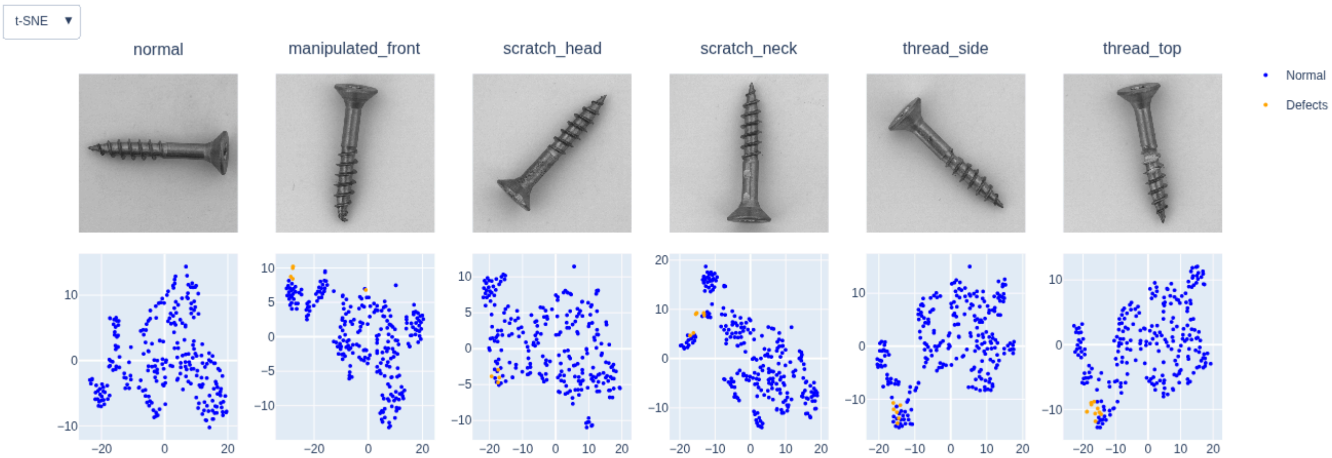


Figure 9: Example of different types for the Screw category

The patch projections in Figure 9 provide a visual example of the improved local structure observed for screw in the preceding statistical analysis. Rather than treating defective screw samples as a single group, the projections show how individual defect types are represented relative to the normal patches within each image For manipulated_front, where the tip of the screw is defective, most anomalous patches form a relatively localised region near the edge of the normal patch distribution, although one anomalous patch lies much closer to the normal patch structure. This partial overlap is qualitatively consistent with the comparatively low Calinski-Harabasz Score reported for screw in Table 5a, since the defective patches are not uniformly separated from the surrounding normal representation.

The scratch and thread defects exhibit somewhat different behaviour. In these images, the anomalous region spans a slightly larger number of patches, which generally form more compact local groups relative to the surrounding normal patches. This is particularly visible for scratch_neck, thread_side, and thread_top. The greater local compactness of these defective patch groups is qualitatively consistent with the lower Davies-Bouldin value observed for screw at the patch level in Table 5a, indicating lower within-group dispersion relative to the separation between normal and anomalous patches.

These examples help explain the contrast between the patch and CLS results discussed previously. Screw anomalies can produce locally distinctive patch representations even when only a small number of patches are affected. When these local representations contribute to the global CLS token, however, their influence may be small relative to the much larger amount of normal screw information present in the image. The patch projections therefore provide qualitative support for the earlier observation that locally coherent anomaly information can be present even where normal and defective CLS embeddings remain difficult to distinguish.

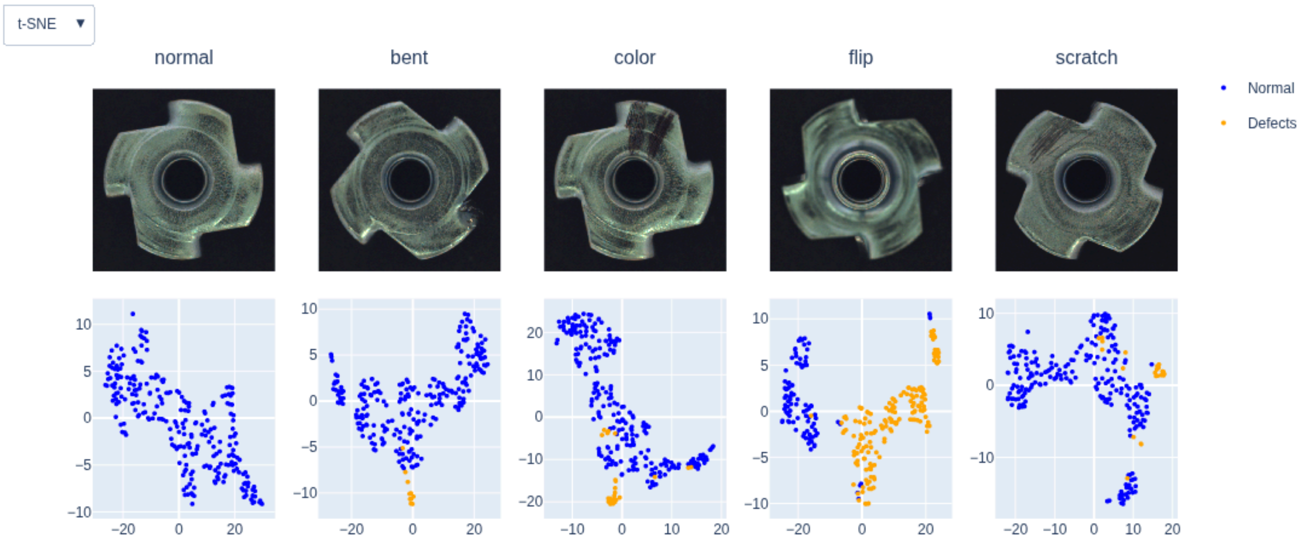


Figure 10: Example of different types for the Metal-Nut category

The patch projections in Figure 10 provide an example of how the number and distribution of anomalous patches may influence the category-level statistics observed previously. The flip defect is particularly notable, as defective patches occupy a substantially larger proportion of the projected space than for the other defect types. Despite this large anomalous region, many of these patches remain close to the normal patch distribution, which may contribute to the Separation Ratio of approximately 1.00 (0.9997) observed in Table 5b. This is consistent with the earlier observation that a spatially large structural defect does not necessarily produce locally distinctive patch representations. Furthermore, flip defects account for approximately 20% of the defective metal nut test samples in Figure 4, meaning that their patch-level behaviour may have a meaningful influence on the category-level statistic. However, because the reported Separation Ratio aggregates all defect types, the contribution of flip cannot be isolated from the current analysis. Computing the metric separately for each defect type would be required to determine how strongly this behaviour contributes to the overall metal nut result.

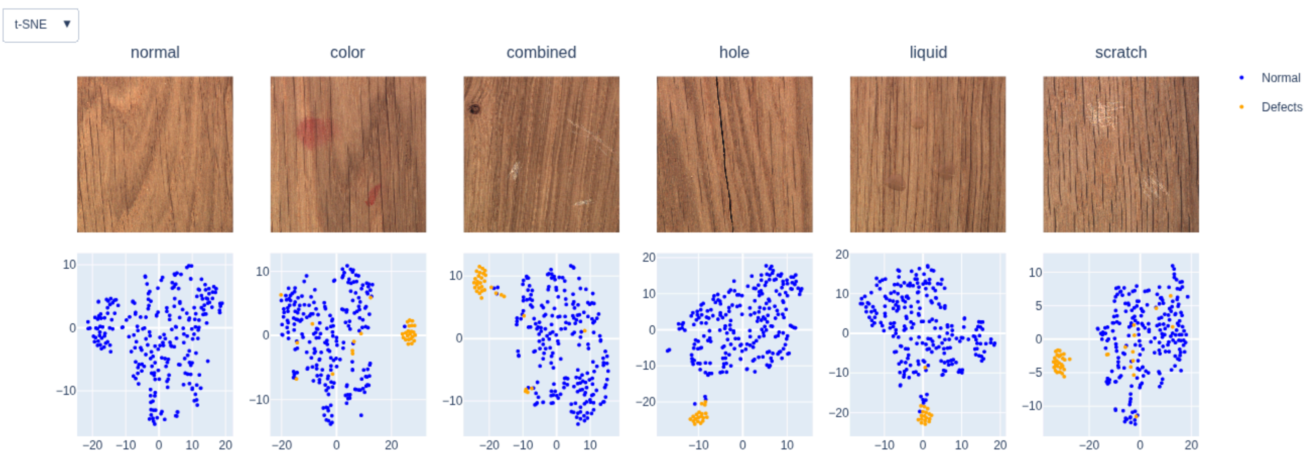


Figure 11: Example of different types for the Wood category

The patch projections in Figure 11 provide a possible explanation for the comparatively poor Normal Test/Train retention ratio observed in Table 5c. Wood achieves the lowest ratio of all categories at 0.8228, with the Normal Intra-Cluster Distance decreasing from 28.6674 for the training patches to 23.5885 for the normal patches within defective test images. Although wood has a relatively uniform overall appearance, its local texture contains substantial variation in the orientation, spacing, and contrast of the grain. Different grain patterns may therefore cause normal patches to occupy relatively dispersed regions of the embedding space, while differences in the grain represented within the training and test samples could contribute to the relatively large change in compactness between the two distributions. This behaviour can also be linked back to the CLS statistics in Table 4, where wood exhibits the poorest Normal Inter-Cluster result and the highest Normal Intra-Cluster Distance (18.9459). Together, the poor patch retention ratio and comparatively dispersed CLS representation suggest that natural variation in wood grain may affect the stability of both the local patch embeddings and the resulting global representation.

Despite this variation within the normal representation, wood still achieves a relatively high Mahalanobis AUROC of 0.944 in Table 3. This may be partly explained by the structure of its defective patch embeddings. Wood achieves a comparatively low patch-level Davies-Bouldin Index of 1.4002 and a high Calinski-Harabasz Index of 40.1426, together with a Separation Ratio of 1.4054, indicating that its anomalous patches generally remain well separated from the normal reference while forming relatively coherent local groups. The projections in Figure 11 support this interpretation, with the different wood defect types generally producing compact anomalous regions that remain distinguishable from the surrounding normal patches. Therefore, although natural variation in wood grain may reduce the stability of the normal representation between training and test samples, the defects themselves can still introduce sufficiently distinctive local changes to support strong anomaly-detection performance.

4. Correlation

This section examines relationships between the dataset characteristics and embedding-space statistics discussed in the preceding sections. The analysis considers relationships ranging from dataset properties, such as defect coverage, and anomaly-detection performance, to associations between patch- and CLS-level embedding geometry. The aim is to identify which characteristics are associated with stronger anomaly separation and to investigate how local patch structure relates to the final global representation.

A recurring consideration throughout the correlation analysis is the influence of the screw category. Screw represents an unusually difficult category and exhibits several extreme values in the preceding embedding analysis. Consequently, a single observation can have substantial influence on a linear correlation calculated across only 15 categories. Correlations are therefore examined both across the complete dataset and, where relevant, with screw excluded as a sensitivity analysis. The complete analysis represents the behaviour of all MVTec AD categories, while the screw-excluded analysis is used to determine whether broader relationships between the remaining categories are being obscured or disproportionately influenced by this extreme case. Screw is not treated as invalid data; rather, the comparison allows its influence on the observed relationships to be examined explicitly.

4.1. Defects Size vs AUROC

As proposed in Section 3, one objective of the embedding analysis was to investigate whether the spatial extent of a defect is associated with anomaly-detection performance. Several observations throughout the preceding analysis suggested that defect coverage alone had limited explanatory power; however, the relationship can be examined directly by comparing average category-level defect coverage with AUROC.

As shown in Figure 12, the observed relationship is substantially weaker than initially expected. It was hypothesised that larger anomalous regions might be easier for the pretrained representation to distinguish, since they alter a greater proportion of the image and therefore have greater potential to influence the resulting global representation. However, little evidence of such a relationship is observed across the categories. Screw represents an influential case, combining both the smallest average defect coverage and the lowest AUROC. When screw is excluded, the already weak relationship is reduced further, with defect coverage exhibiting effectively no linear association with AUROC.

This suggests that the spatial extent of an anomaly alone is not a strong determinant of category-level anomaly-detection performance. Instead, characteristics such as the visual distinctiveness of the defect and the resulting embedding-space geometry are likely to be more informative. Defect coverage may still influence individual categories, particularly in extreme cases such as screw, but it does not appear to provide a general explanation for differences in AUROC across MVTec AD.

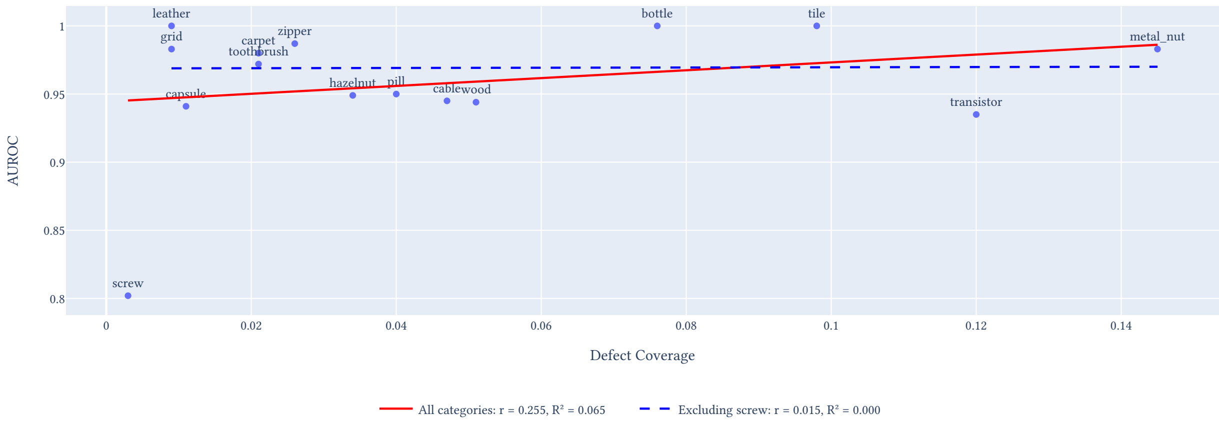


Figure 12: AUROC vs Defect Coverage with fitted trendlines

4.2. CLS Correlation

This section examines how CLS-level embedding statistics relate to the final AUROC and whether particular characteristics of the global embedding space are more strongly associated with anomaly-detection performance.

As the analysis is performed across the 15 MVTec AD categories, the correlations are treated as exploratory associations rather than evidence of causal relationships. Sensitivity analyses are additionally performed where individual categories exhibit extreme metric values to determine how strongly these observations influence the resulting correlations.

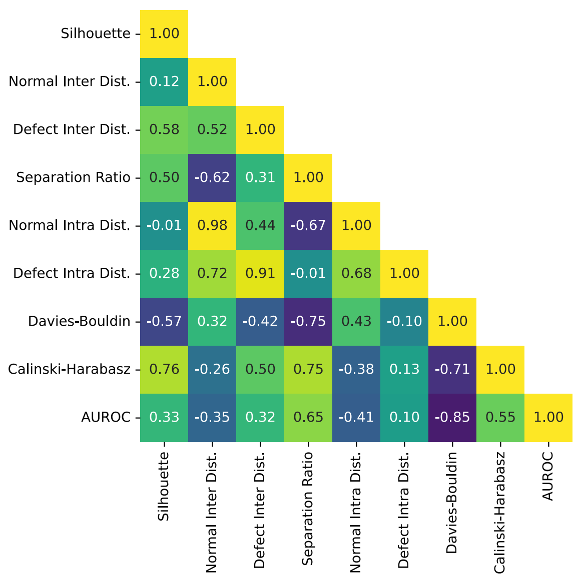


Figure 13: CLS metric correlations across all categories

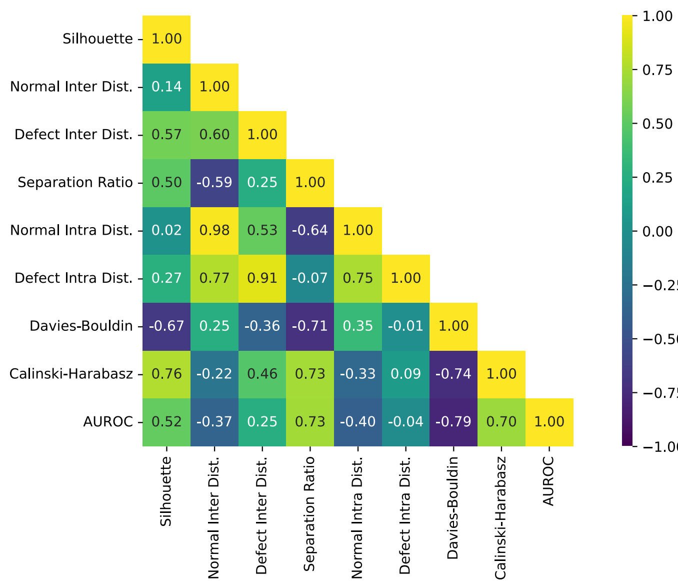


Figure 14: CLS metric correlations with screw excluded

At the CLS level, the Davies-Bouldin Index exhibits the strongest relationship with AUROC, with a strong negative correlation of approximately $r = -0.85$. As shown in Figure 15, the corresponding R^2 of approximately 0.716 indicates that 71.6% of the observed category-level variation in AUROC is associated with the linear relationship with DB. This suggests that the balance between normal-defective separation and within-group dispersion captured by DB is strongly associated with anomaly-detection performance. Table 4 reports a mean DB of 3.28 across categories, with a standard deviation of 1.4754 and values ranging from 1.5058 for leather to 6.6545 for screw. This substantial variation between categories appears particularly informative when considered alongside their anomaly-detection performance.

The strength of this relationship is partly influenced by the screw category. Screw exhibits both the highest DB value and the lowest AUROC among the evaluated categories, placing it at an extreme of the observed relationship. When screw is excluded from the analysis in Figure 14, the correlation between DB and AUROC weakens from $r = -0.85$ to $r = -0.79$, with the corresponding R^2 decreasing from approximately 0.716 to 0.624. Therefore, although screw strengthens the observed relationship, DB remains strongly associated with AUROC across the remaining categories, accounting for approximately 62.4% of the observed variation in AUROC under the linear model. Interestingly, several other CLS statistics exhibit stronger correlations with AUROC after screw is removed. This suggests that screw is not simply a noisy observation, but an unusual category whose embedding structure is captured particularly strongly by DB while differing from broader relationships observed across the remaining categories.

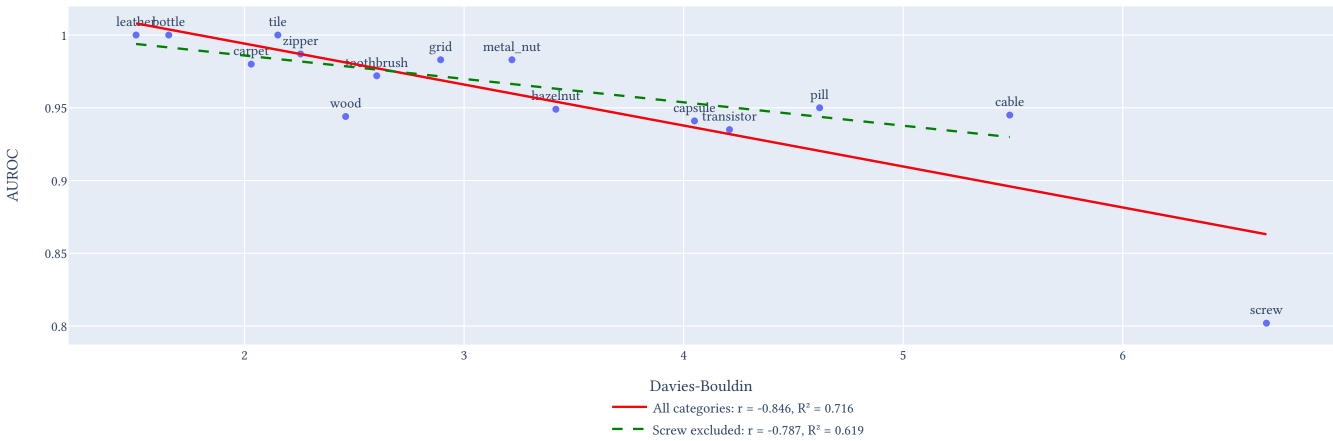


Figure 15: Davies-Bouldin Index vs AUROC

The Calinski-Harabasz Index provides a complementary view of this relationship by considering normal-defective separation relative to the variation within the two groups. Unlike DB, higher CH values indicate stronger cluster structure [REF]. When screw is excluded, as shown in Figure 14, CH exhibits a moderately strong positive correlation with AUROC ($r = 0.70$), corresponding to an R^2 of 0.49. This indicates that approximately 49% of the observed variation in category-level AUROC is associated with the linear relationship with CH. The result therefore provides additional evidence that anomaly-detection performance is associated not simply with the absolute displacement of defective embeddings, but with how strongly the normal and defective groups are separated relative to their internal variability.

Table 4 reports a mean CH of 8.8111 with a standard deviation of 6.7326, ranging from 2.6555 for screw to 27.5506 for leather. This large variation is partly influenced by the extreme values of screw and leather at opposite ends of the distribution. As shown in Figure 16, categories with higher CH values generally tend to achieve higher AUROC, supporting the observed positive correlation. However, the relationship is not deterministic. Leather combines the highest CH with an AUROC of 1.0, while bottle and tile also achieve an AUROC of 1.0 despite substantially lower CH values. Similarly, metal nut and grid achieve AUROC values close to 1.0 with CH values considerably below that of leather and closer to the lower end of the observed range. This suggests that strong normal-defective cluster structure is associated with improved anomaly-detection performance, but represents only one of several characteristics contributing to category-level AUROC.

When both extreme categories, screw and leather, are excluded, the relationship strengthens slightly further in Figure 16 to $r = 0.716$, with $R^2 = 0.513$. Therefore, approximately 51.3% of the observed AUROC variation among the remaining categories is associated with the linear relationship with

CH. Importantly, the correlation does not weaken when the two extreme CH observations are removed, suggesting that the relationship is not dependent on these categories alone. CH may therefore provide a useful diagnostic metric when evaluating subsequent fine-tuning or modifications to the embedding space, particularly for assessing whether changes to normal-defective cluster structure are associated with changes in anomaly-detection performance.

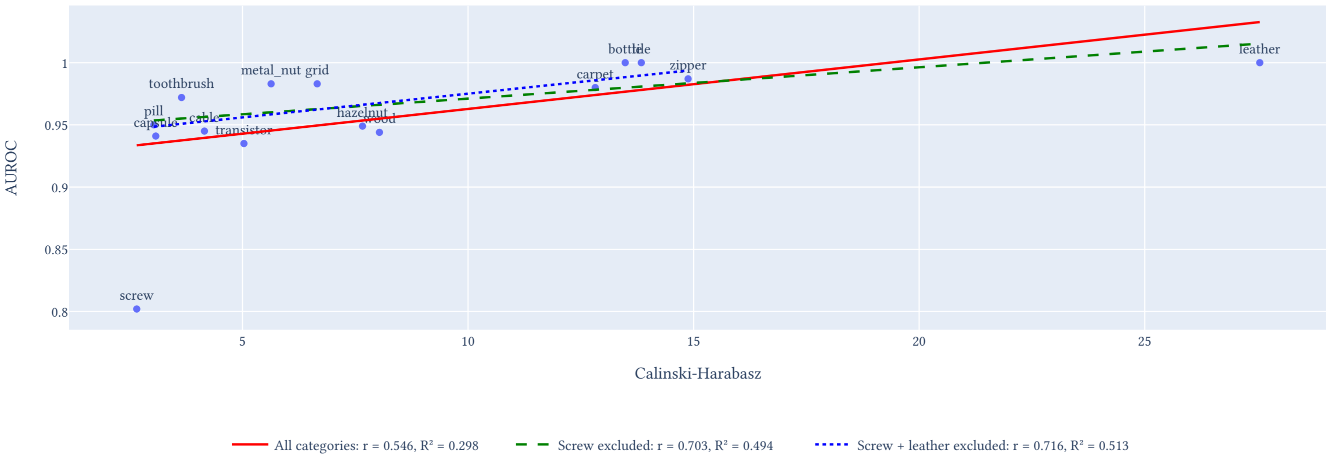


Figure 16: Calinksi-Harabasz Index vs AUROC

The last notable, although less surprising, relationship is between Separation Ratio and AUROC. This relationship becomes particularly clear in Figure 14 when screw is excluded. As observed previously in Figure 8, screw contains substantial overlap between normal and defective CLS embeddings and represents one of the most diffuse categories. More generally, categories with a larger Separation Ratio tend to achieve higher AUROC, which is consistent with the intended behaviour of the anomaly representation: as defective embeddings move farther from the learned normal reference relative to normal test embeddings, they become easier to distinguish using distance-based anomaly scoring. Unlike the relationship observed between DB and AUROC, the positive association between Separation Ratio and AUROC is largely expected from the definition of the metric and therefore serves primarily as confirmation that the observed embedding geometry is reflected in anomaly-detection performance.

4.3. Patch Correlation

This section examines how patch-level embedding statistics relate to the final image-level AUROC and whether particular characteristics of the local embedding space are associated with anomaly-detection performance. Unlike the preceding CLS analysis, these statistics describe local patch behaviour rather than the final global image representation. The correlations are therefore used to investigate whether properties of the local representation are reflected in the performance achieved using the CLS embeddings.

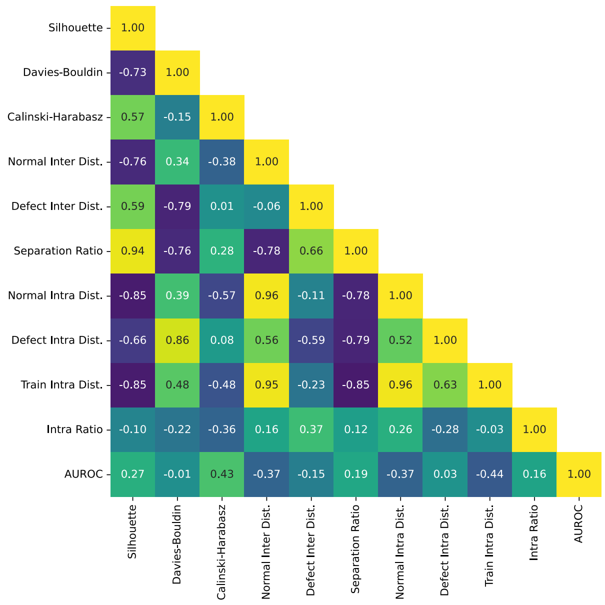


Figure 17: Patch metric correlations across all categories

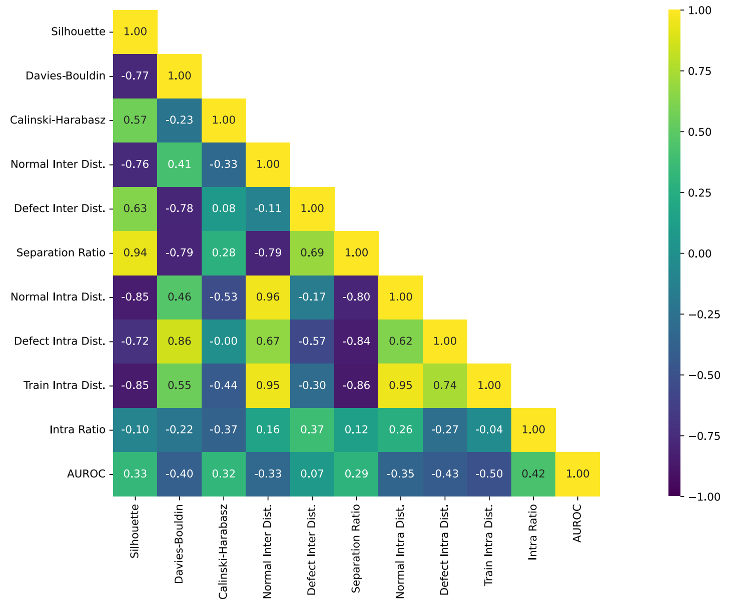


Figure 18: Patch metric correlations with screw excluded

At the patch level, the Davies-Bouldin Index exhibits substantially different behaviour from the corresponding CLS metric. When all categories are considered in Figure 17, patch DB has almost no correlation with AUROC ($r = -0.01$). However, when screw is excluded in Figure 18, the correlation increases in magnitude to approximately $r = -0.40$. This remains considerably weaker than the relationship observed at the CLS level, although the two metrics describe different structures and are therefore not directly comparable. The patch DB statistic is calculated from normal and anomalous patches within individual defective images before being aggregated across the category. Categories with very small defect regions, such as screw with an average coverage of approximately 0.3% in Figure 6, contain relatively few anomalous patches within each image. This may make category-level patch statistics more sensitive to the behaviour of a small number of anomalous regions and contribute to greater variability in their relationship with final image-level AUROC.

A notable relationship is observed between Normal Train Intra-Cluster Distance and AUROC, with a moderate negative correlation of approximately $r = -0.44$. When screw is excluded, this strengthens to approximately $r = -0.50$. Although screw has a relatively large training intra-cluster distance and therefore lies in the expected direction of the negative relationship, its AUROC is substantially lower than would be expected from the broader linear trend, making it an influential category in the analysis. Its removal therefore produces a more consistent relationship across the remaining categories. The strengthened relationship among the remaining categories provides some evidence that compactness of the normal training patch representation may be a useful diagnostic during future fine-tuning experiments. However, further experiments would be required to determine whether deliberately increasing compactness leads to improved AUROC.

Another interesting patch-level relationship is observed between Defect Intra-Cluster Distance and AUROC. When screw is excluded in Figure 18, the metric exhibits a negative correlation of $r = -0.43$, corresponding to $R^2 = 0.18$. This indicates a weak but notable association in which categories with more compact anomalous patch representations tend to achieve stronger image-level anomaly-detection performance. As discussed previously in page 16, this also raises the question of how local patch structure is aggregated into the final CLS representation. If anomalous patches form coherent regions within the embedding space, this structure may contribute useful information to the global representation. However, Defect Intra-Cluster Distance measures only the internal compactness of the anomalous patches and does not describe how those patches are positioned relative to the normal representation.

Importantly, Defect Intra-Cluster Distance alone does not determine AUROC. For example, from Table 5c, wood has a lower Defect Intra-Cluster Distance of 20.31 than metal nut at 25.82, indicating a more compact anomalous patch representation, yet achieves a lower AUROC of 0.944 compared with 0.983 for metal nut in Table 3. This demonstrates that compactness of anomalous patches is only one component of useful anomaly structure. A compact defect cluster may still occupy a region close to, or enclosed by, the normal patch distribution, while a more dispersed defect representation may remain sufficiently distinct to support strong detection. Future work could therefore examine Defect Intra-Cluster Distance together with measures of normal-defect separation and investigate how coherent but discriminative anomalous patch structure is aggregated into the CLS representation. In a semi-supervised setting, this could additionally motivate experiments examining whether explicitly encouraging such structure improves the resulting CLS representation and final AUROC.

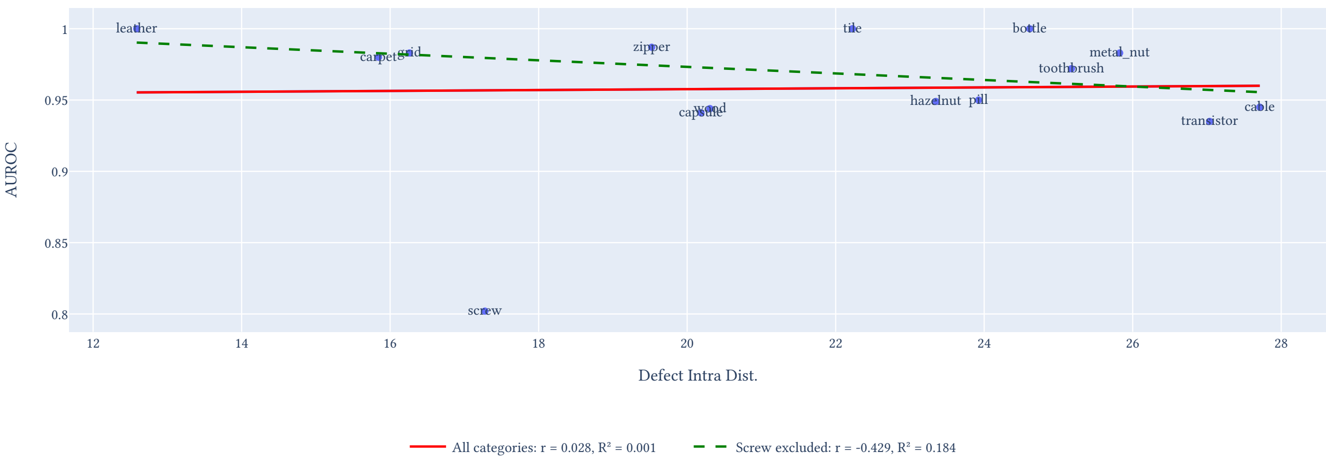


Figure 19: AUROC vs Defect Intra Distance

4.4. CLS ↔ Patch Correlation

This section examines relationships between CLS- and patch-level embedding statistics to determine whether properties of the local patch representation are reflected in the final global representation. These correlations do not directly describe how patch information is aggregated into the CLS token, but may identify local embedding characteristics associated with desirable CLS-level properties. In particular, this analysis considers whether patch-level characteristics could provide useful objectives or diagnostics during future fine-tuning, with the aim of improving CLS metrics such as the Davies-Bouldin Index and Calinski-Harabasz Index.

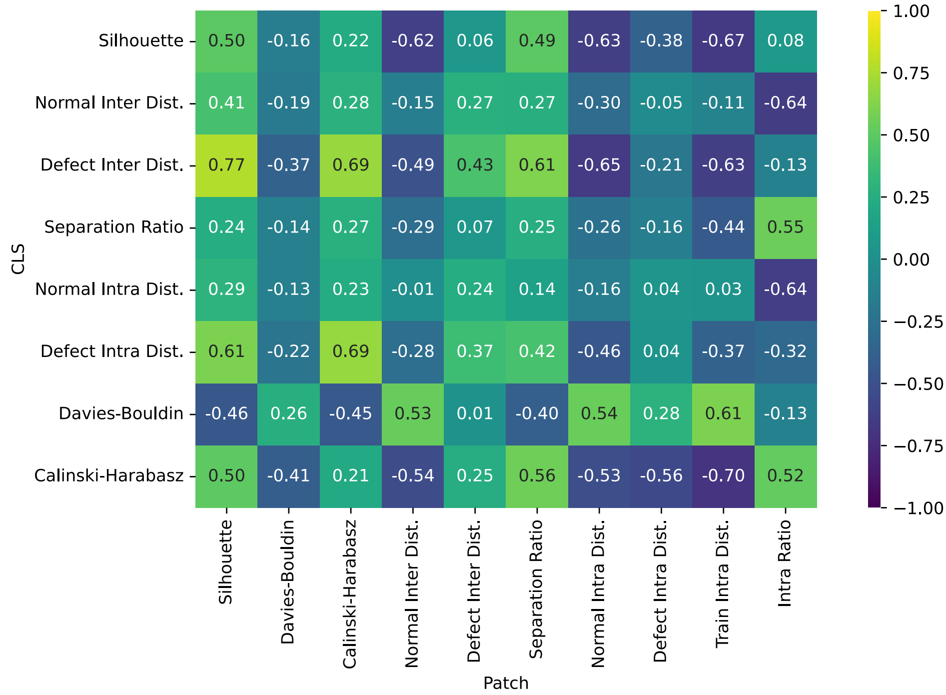


Figure 20: Correlation matrix between CLS and Patch metrics

The patch-level metric most strongly associated with the CLS metrics previously linked to anomaly-detection performance, the Davies-Bouldin Index and Calinski-Harabasz Index, is Patch Train Intra-Cluster Distance. As shown in Figure 20, Train Intra Distance has correlations of $r_{CH} = -0.70$ with CLS Calinski-Harabasz and $r_{DB} = 0.61$ with CLS Davies-Bouldin. Both relationships are consistent in direction: more compact normal training patch representations, represented by a lower Train Intra Distance, tend to be associated with higher CLS Calinski-Harabasz values and lower CLS Davies-Bouldin values. This suggests that the compactness of the normal patch representation may be related to the quality of the resulting global normal-defective structure.

The relationship with CLS Davies-Bouldin is shown in Figure 21. Although screw lies somewhat outside the main concentration of categories, excluding it has almost no effect on the strength of the linear relationship, with R^2_{DB} changing only from 0.366 to 0.369. This indicates that the relationship is not dependent on the unusual behaviour of screw and instead reflects a moderate trend across the remaining categories.

A similar relationship is observed for CLS Calinski-Harabasz in Figure 22. Leather represents a more extreme observation in this relationship, and its removal reduces R^2_{CH} from 0.493 to 0.377. Leather therefore contributes more strongly to the observed relationship than screw does in the Davies-Bouldin comparison. However, the relationship remains present after its removal, suggesting that it is not solely produced by this category. Taken together, these results indicate that categories with more compact normal training patch representations tend to exhibit more favourable CLS cluster structure. This is particularly relevant for future fine-tuning because Patch Train Intra-Cluster Distance can be measured using only the normal training representation, making it a potentially useful unsupervised diagnostic for investigating whether changes in local patch compactness are accompanied by improvements in CLS Davies-Bouldin and Calinski-Harabasz.

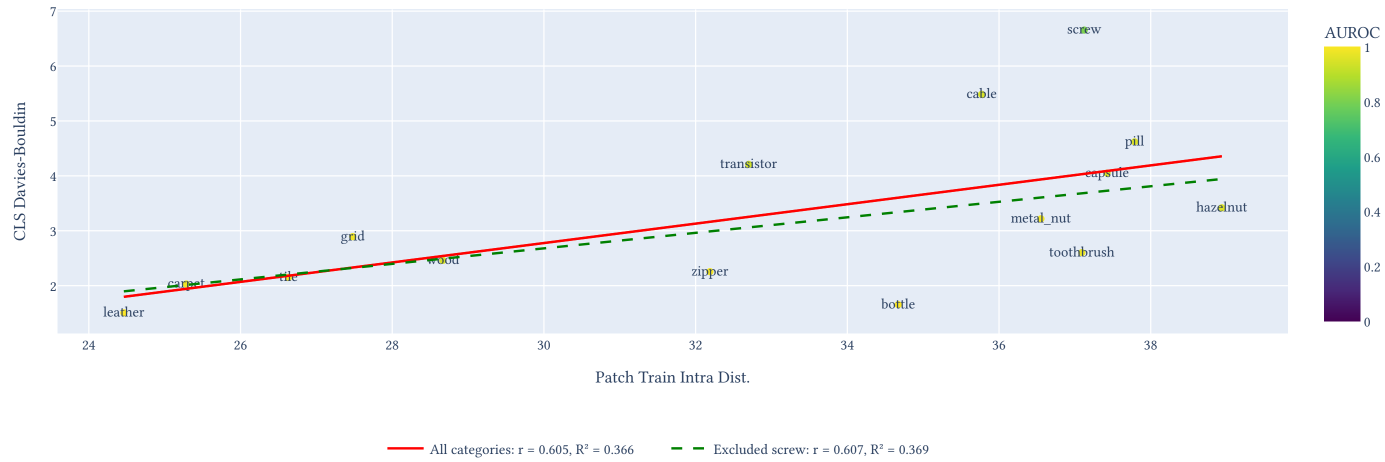


Figure 21: CLS Davies-Bouldin vs Patch Train Intra Dist.

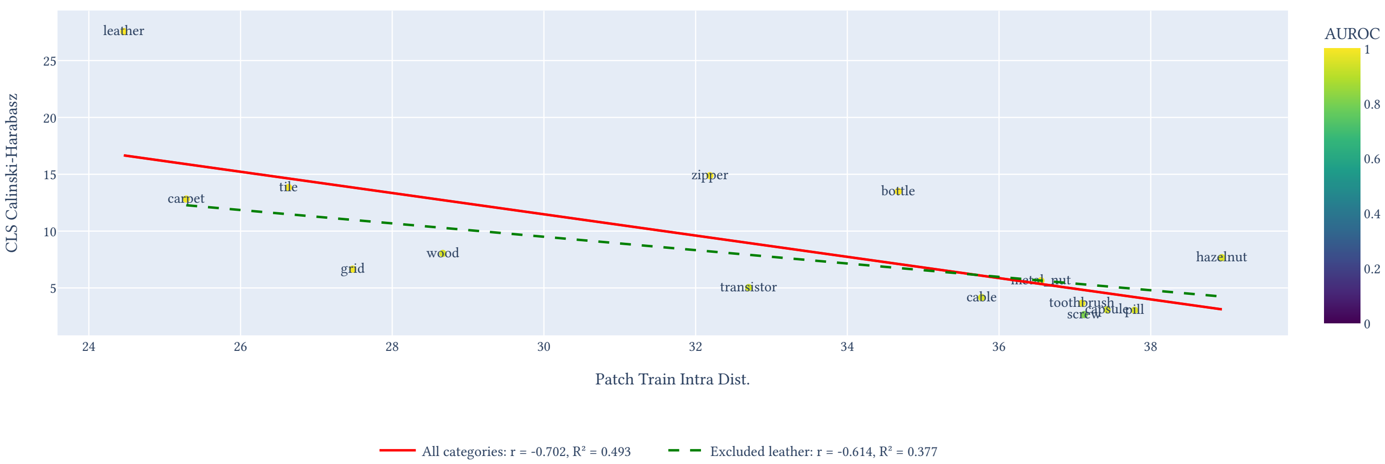


Figure 22: CLS Calinski-Harabasz vs Patch Train Intra Dist.

The strongest observed relationship between the CLS- and patch-level metrics is between CLS Defect Inter-Cluster Distance and Patch Silhouette, with $r = 0.77$. As shown in Figure 23, the corresponding linear fit has an R^2 of 0.587, indicating that approximately 58.7% of the observed category-level variation in CLS Defect Inter-Cluster Distance is associated with its linear relationship with Patch Silhouette. This suggests that categories in which anomalous patches are more clearly separated from normal patches also tend to place their global defective representations further from the learned normal reference.

The distribution of categories along this relationship is also notable. Several visually more uniform or structurally simpler categories identified during the earlier EDA occupy the upper region of the plot, while the more heterogeneous object categories are concentrated at lower Patch Silhouette values. This initially suggests that the overall correlation may partly reflect systematic differences between these category groups rather than a direct relationship between patch separation and CLS aggregation alone. However, the separate trend lines in Figure 23 show that the relationship remains within both groups. The object categories exhibit the stronger within-group relationship at $r = 0.540$ and $R^2 = 0.292$, compared with $r = 0.440$ and $R^2 = 0.194$ for the more uniform categories. The overall relationship is therefore strengthened by the separation between the two groups, but is not solely produced by it. In particular, the continued relationship across the more visually varied object categories provides evidence that local normal-defect separation and global defective

displacement are related properties of the pretrained representation rather than simply consequences of category uniformity.
This relationship is therefore particularly relevant for future fine-tuning experiments. If improvements in patch-level normal-defect separation are consistently accompanied by greater CLS-level defective displacement, Patch Silhouette or related local separation metrics could potentially be monitored or incorporated into a patch-level objective. The object-level trend is especially relevant here, as it suggests that this relationship persists even across categories with substantially greater visual and structural variation. However, the observed correlations do not establish that improving patch separation will directly improve CLS geometry. Determining whether such a relationship is causal would require future experiments that track both patch- and CLS-level metrics throughout fine-tuning and examine whether deliberate changes to the local representation produce corresponding changes in the global representation.

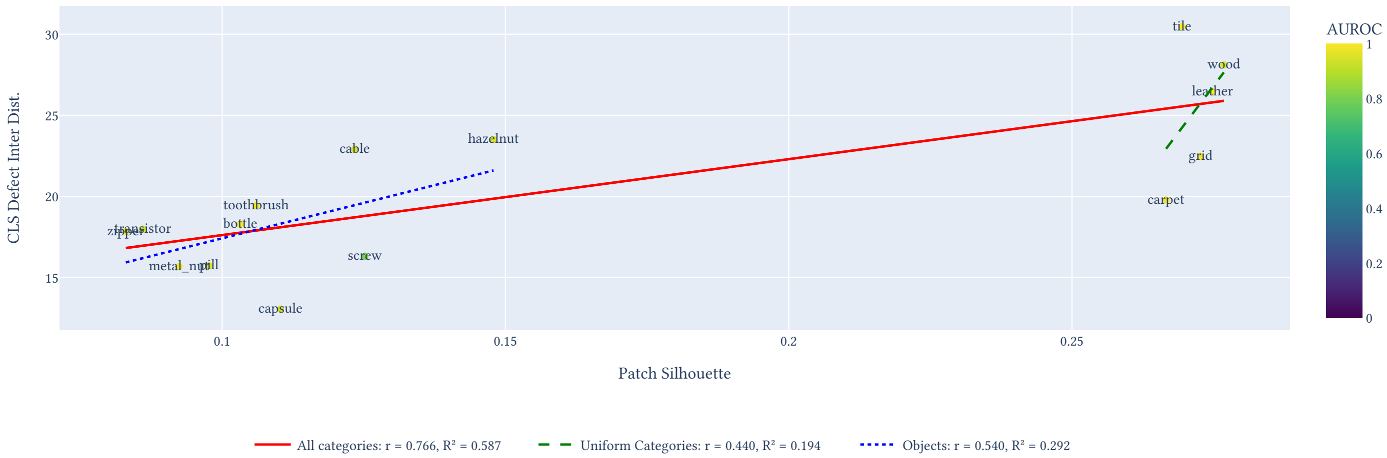


Figure 23: CLS Defect Inter Dist. vs Patch Silhouette

Another notable relationship is observed between CLS Separation Ratio and Patch Normal Test/Train Intra Ratio, with a Pearson correlation of $r = 0.55$. A weaker relationship is also observed between CLS Separation Ratio and Patch Train Intra-Cluster Distance at $r = -0.44$, while the corresponding relationship with Patch Normal Test Intra-Cluster Distance is weaker again at $r = -0.26$. Together, these results suggest that the relationship between the normal training and test patch distributions may contain more information about global separation than either of the individual intra-cluster distances alone.

However, the distribution of categories does not follow a particularly clear linear structure, with most categories concentrated around Test/Train ratios close to 1. A Spearman rank correlation was therefore additionally calculated to determine whether the relationship was better described as monotonic rather than linear. This produces a substantially stronger correlation of $\rho = 0.75$, compared with the Pearson correlation of $r = 0.55$. When wood and leather, which occupy the extremes of the Patch Intra Ratio, are excluded, the Pearson correlation decreases to $r = 0.48$, while the Spearman correlation remains similar at $\rho = 0.72$. This suggests that the association is not solely produced by these extreme categories, but that higher Patch Intra Ratios generally correspond to higher CLS Separation Ratios in a relationship that is not well represented by a single straight line.

Because a Test/Train Intra Ratio close to 1 represents similar normal-patch dispersion between the training and test distributions, the absolute deviation of the ratio from 1 was also examined. This produces only a weak linear relationship with CLS Separation Ratio at $r = -0.252$, but a stronger negative Spearman correlation of $\rho = -0.504$. The direction of this relationship is consistent with categories whose normal patch dispersion remains closer to its training behaviour tending to exhibit greater CLS normal-defect separation. However, the weaker relationship compared with the raw Intra Ratio shows that preservation of normal-patch dispersion alone does not explain the stronger monotonic association observed previously.

The category-level results further demonstrate this limitation. As shown in Table 5c, wood has the lowest Patch Intra Ratio at 0.8228, indicating substantially more compact normal test patches relative to its normal training representation, and does not exhibit correspondingly strong CLS separation. Leather instead has a ratio of 1.0831, indicating greater normal test-patch dispersion than in its training representation, while bottle has a ratio close to 1 at 0.9763 yet achieves the strongest CLS Separation Ratio. These differences indicate that stability of normal patch dispersion may be one contributing property of the representation, but cannot independently determine global separation. Category-specific visual structure, the characteristics of the anomalous regions, and the way local anomalous information is aggregated into the CLS representation are also likely to contribute.

These observations are particularly relevant when considering future fine-tuning objectives. The relationships identified here suggest that useful patch representations should not necessarily be reduced to a simple objective of maximising normal-defect separation or compressing each group into a single compact region. When a defective image is processed, preserving meaningful local variation across its patch representations may provide the CLS token with a richer representation of both the underlying normal structure and the anomalous regions. The Test/Train Intra Ratio results suggest that substantial contraction of the normal patches within defective images is associated with weaker CLS separation, while the Defect Intra-Cluster Distance results also demonstrate that strong anomaly-detection performance does not require defective patches to collapse into a single compact region. Together, these observations suggest that the geometry retained across the patch representation may be more important than simple class compactness alone. Future fine-tuning experiments should therefore track how patch-level geometry changes alongside CLS Davies-Bouldin, Calinski-Harabasz, Separation Ratio, and AUROC, rather than assuming that a generic contrastive objective such as InfoNCE will improve the representation simply by increasing separation.

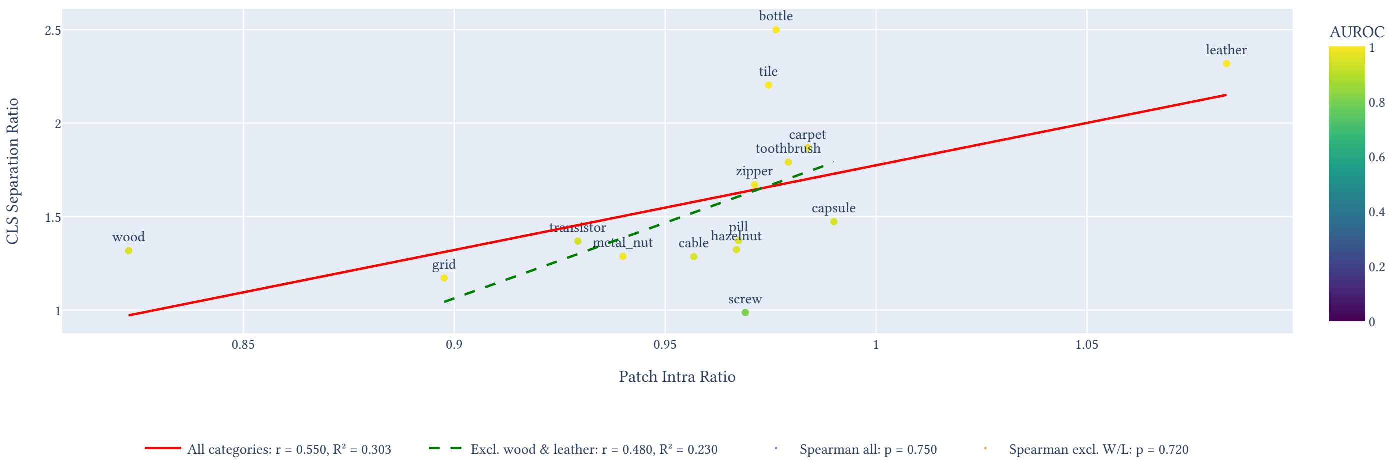


Figure 24: CLS Separation Ratio vs Patch Intra Ratio

Bibliography

[1] M. Software, “The MVTec Anomaly Detection Dataset,” [Online]. Available: https://www.mvtec.com/research-teaching/datasets/mvtec-ad